

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

November 2013

EC207 Control Systems

Date: 13.11.2013, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

## SECTION - I

Q - 1 (a) Explain classification of control systems [04]

(b) State the differences between open loop control system and close loop control system. [03]

Q - 2 (a) Obtain the transfer functions  $\frac{X_1(s)}{F(s)}$  and  $\frac{X_2(s)}{F(s)}$  of the mechanical system shown in figure 1. [7]

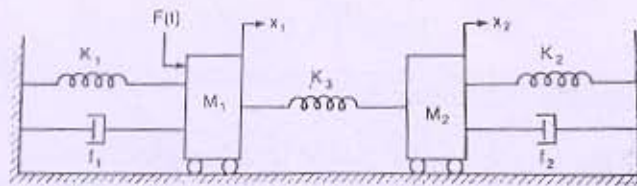


Figure 1.

(b) For the system represented by the block diagram shown in figure 2, evaluate the close loop transfer function. [7]

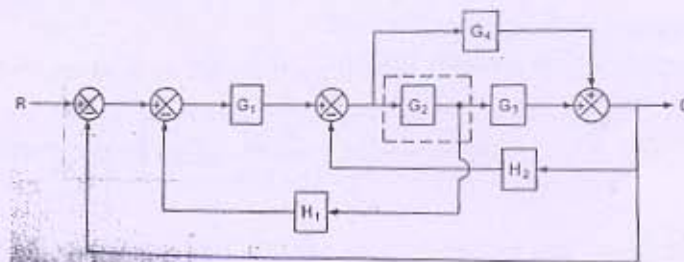


Figure 2.

OR

Q - 2 (a) Write the differential equations for the mechanical system shown in figure 3. [7]

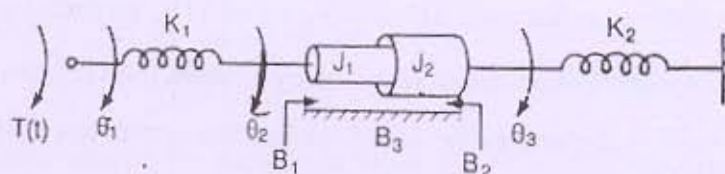


Figure 3.

- (b) For the signal flow graph shown in figure 4, find  $\frac{C}{R}$ . [7]

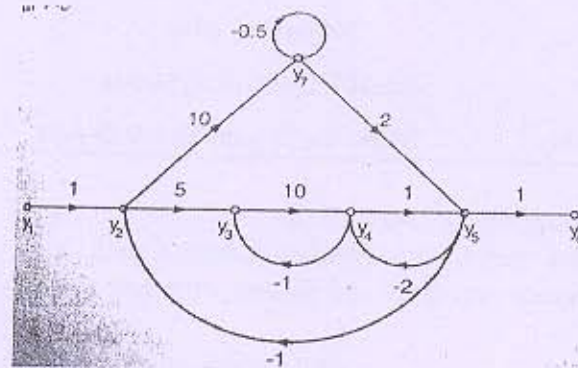


Figure 4.

- Q - 3 (a) State the differences between block diagram reduction technique and signal flow graph technique. [04]
- (b) State the differences between field controlled DC servomotor and armature controlled DC servomotor. [04]
- (c) On the basis of certain measurements, the system shows following response [06]

$$C(t) = 1 + 0.2 e^{-60t} - 1.2 e^{-10t}$$

When subjected to a unit-step input

- (i) Find the expression for closed loop transfer function  
(ii) Obtain the undamped natural frequency and damping ratio of the system.

OR

- Q - 3 (a) Determine the sensitivity of the overall (closed-loop) transfer function for the system in figure 5 at  $\omega = 1$  rad/sec. W.r.t. [04]
- a) Forward path transfer function  
b) Feedback path transfer function

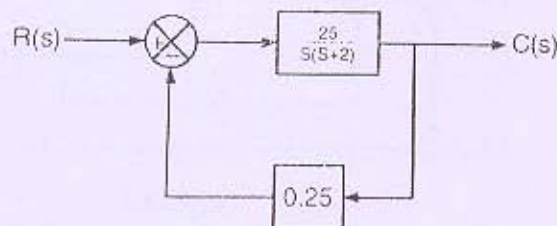


Figure 5.

- (b) State the differences between AC servomotor and DC servomotor [04]
- (c) A unity feedback system is characterised by an open loop transfer function [06]
- $$G(s) = \frac{K}{s(s+10)}$$
- Determine the gain K so that the system have a damping ratio of 0.5. For the value K determine settling time, peak overshoot and time to peak overshoot for a unit step input



Section II

Q - 4 (a) Give definitions of the following: [04]

- 1) Step function input signal
- 2) Ramp function input signal
- 3) Rise time ( $t_r$ )
- 4) Delay time ( $t_d$ )

(b) Explain the advantages of state variable. [03]

Q - 5 (a) The characteristic equation of a system is given by [07]

$$s^4 + 8s^3 + 18s^2 + 16s + 4 = 0$$

Determine the stability of a system by means of Hurwitz criterion.

(b) A feedback control system has n open loop transfer function [07]

$$G(s)H(s) = \frac{K(s+2)}{(s+1)^2}$$

Find the root locus as K is varied from 0 to  $\infty$ .

**OR**

Q - 5 (a) Consider the characteristic equation of a closed loop control system as [07]

$$s^3 + 3Ks^2 + (K+2)s + 4 = 0$$

Find the range of K so that the system is stable

(b) A feedback control system has n open loop transfer function [07]

$$G(s)H(s) = \frac{K}{s(s+3)(s^2+2s+2)}$$

Find the root locus as K is varied from 0 to  $\infty$ .

Q - 6 (a) Sketch the polar plot of  $G(s) = \frac{10}{s(s+1)}$ . [07]

(b) Sketch the bode plot and determine the gain crossover and phase crossover frequencies for the transfer function  $G(s) = \frac{10}{s(1+0.5s)(1+0.1s)}$ . [07]

**OR**

Q - 6 (a) Draw the polar plot of the transfer function  $G(s) = \frac{1}{s^2(1+s)(1+2s)}$ . [07]

(b) Sketch the bode plot for the transfer function  $G(s) = \frac{Ks^2}{(1+0.2s)(1+0.02s)}$ . [07]